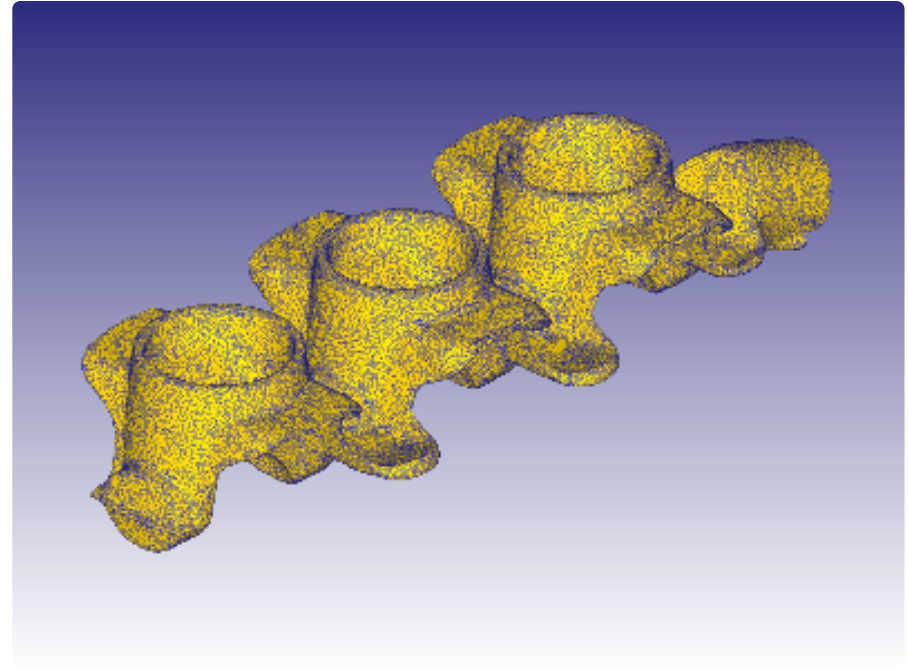


Software and Methods for Forming Process Simulation

Lecture 2 · Modelling and Simulation of Processes I – Forming (MSPO2_6I)

Ing. Martin Necpal, PhD.

WINTER SEMESTER 2026/2027
FACULTY OF MATERIALS SCIENCE AND TECHNOLOGY · STU BRATISLAVA



Closed-die forging of a yoke simulated in DEFORM-3D (© SFTC)

How to control the presentation (bar at the bottom)

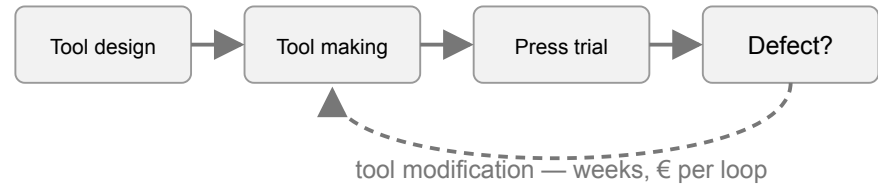
- or the < > buttons – previous / next slide
- Dots – jump directly to a chosen slide
- ► or – play / pause the spoken commentary
- **Auto** on – moves to the next slide when the commentary ends; off – the slide stays
- full screen · slide overview
- + click (Linux: + click) – zoom into a spot on the slide; click again to zoom out
- Underlined abbreviations – hover the mouse over them for an explanation

Motivation Why simulate forming?

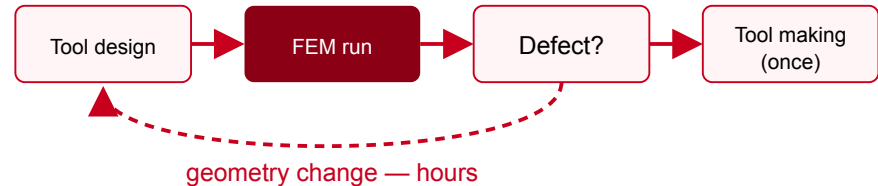
From press-shop trials to virtual try-out

- Forging dies and stamping tools cost **weeks of work and tens of thousands of euros**; every modification adds more
- Most of a product's cost is **fixed at the design stage**, long before production starts
- Simulation moves the iterations **into the computer**: change the preform, die radius or lubricant and run the calculation again
- Result: fewer tool modifications, faster production ramp-up, **less material** (flash, scrap)

Traditional



With simulation



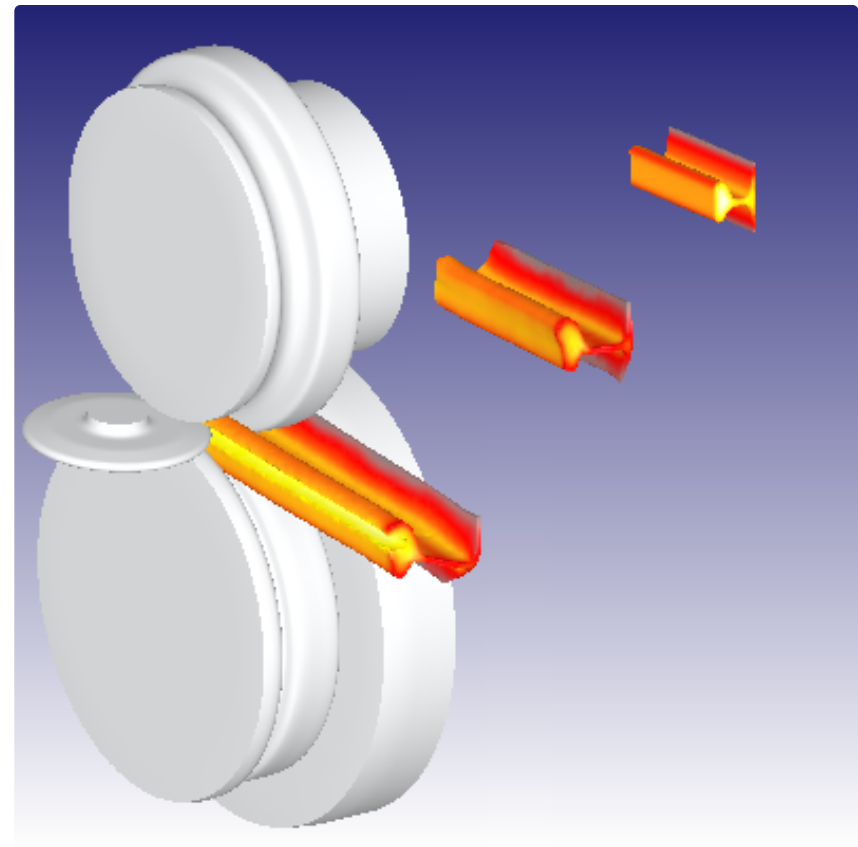
Validation on the press is still needed — but ideally only once.

The iteration loop moves from the press shop into the computer

Motivation What the software tells us

Questions that forming simulation answers

- **Will the die cavity fill?** — underfill, laps, folds, piping
- **What press do I need?** — force and energy versus stroke
- **Will the part crack or neck?** — damage, FLD, thinning
- **Will it keep its shape?** — springback, warpage, distortion
- **What will the tool life be?** — stress, wear, temperature
- **What is inside?** — grain size, phases, hardness



Sequence of several rolling passes in DEFORM-3D (© SFTC)

Where today's lecture fits

1. Real forming process (forging, rolling, stamping ...)



2. Physical model (assumptions, idealisation)



3. **Mathematical model** → analytical solution (Part I)



4. **Numerical model and software** → FEM / FVM (Parts II–IV)



5. Verification and validation (experiment, lectures 5–6)

- **Lecture 1:** modelling, model types, the modelling chain
- **Lecture 2 (today):** methods and **software** that solve the model
- **Lecture 3:** material models — flow stress, hardening
- **Lectures 4–6:** preprocessing, sheet and bulk forming examples
- Lab sessions: DEFORM — so today's Part III is your toolkit for the whole semester

Today's agenda (120 min)

Time	Block	Content
0 – 10	Introduction	Motivation, context, agenda
10 – 25	I · Analytical methods	Slab method, upper bound, slip lines; solving in MATLAB / Python / <u>GAMS</u>
25 – 67	II · Forming software	1 Ansys Forming · 2 Simufact Forming · 3 Abaqus · 4 Moldflow
67 – 97	III · DEFORM	Main part — the software of our lab sessions
97 – 112	IV · Special simulation software	Overview · simple examples · use cases
112 – 120	Summary and questions	Key takeaways, review questions

Part I — Analytical Methods of Forming

Closed-form and semi-analytical models: fast, transparent and the reference for every FEM result

Analytical methods Overview

Classical analytical methods of plasticity theory

Method	Principle	Typical output	Typical process
Slab method (elementary theory)	Force equilibrium on a thin slab; homogeneous deformation	Pressure distribution, force	Upsetting, forging, rolling, drawing
Slip lines	Characteristics of the stress field, rigid-perfectly plastic material, plane strain	Exact stress field, force	Indentation, extrusion, machining
Upper bound (<u>UBM</u> , <u>UBET</u>)	Kinematically admissible velocity field; minimum power	Force $\geq$ actual, die filling	Extrusion, closed-die forging
Lower bound	Statically admissible stress field	Force $\leq$ actual	Limit analysis
Visioplasticity	Measured grid deformation $\rightarrow$ strain rates $\rightarrow$ stresses	Experimental flow field	Extrusion and forging research

- Common assumptions: rigid-plastic material, constant (or mean) flow stress $\bar{\sigma}_f$, simple friction law, simple geometry

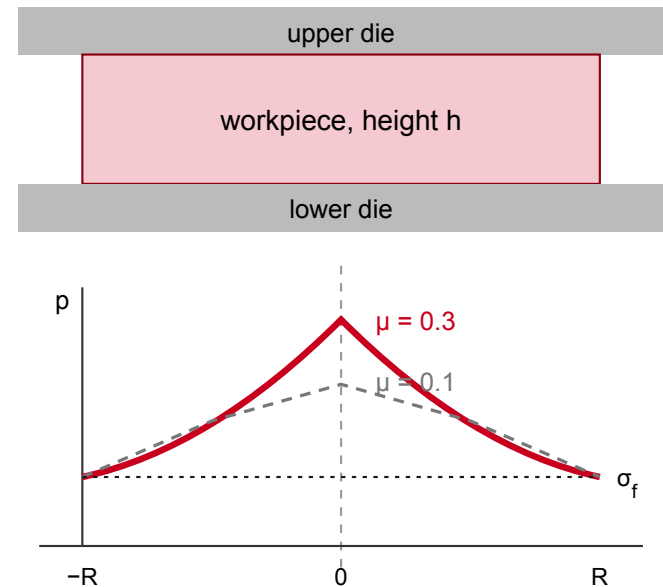
Source: Hosford & Caddell, Metal Forming; Necpal & Sobota, 2023

Slab method: upsetting of a cylinder

$$\frac{dp}{dr} = -\frac{2\mu p}{h} \Rightarrow p(r) = \sigma_f e^{2\mu(R-r)/h}$$

$$F = \int_0^R p(r) 2\pi r dr, \quad R = R_0 \sqrt{h_0/h}$$

- p contact pressure, σ_f flow stress, μ Coulomb friction
- h, R current height and radius (volume constancy)
- Pressure peaks in the centre — the **friction hill**
- Larger μ or smaller $h \Rightarrow$ steeper hill $\Rightarrow$ higher force



Friction hill — contact pressure along the radius (schematic)

Drawing and extrusion: closed-form results

Sachs (slab method) — wire drawing stress

$$\sigma_d = \sigma_f \frac{1+B}{B} \left[1 - \left(\frac{A_1}{A_0} \right)^B \right], \quad B = \mu \cot \alpha$$

- α die semi-angle, A_0, A_1 cross-section before and after
- Condition: $\sigma_d \leq \sigma_f \Rightarrow$ maximum reduction per pass

Upper bound — extrusion pressure

$$\frac{p}{\sigma_f} \approx \underbrace{\ln R_{ext}}_{\text{ideal}} + \underbrace{\mu \cot \alpha \ln R_{ext}}_{\text{friction}} + \underbrace{\frac{2}{\sqrt{3}} \left(\frac{\alpha}{\sin^2 \alpha} - \cot \alpha \right)}_{\text{redundant (shear)}}$$

- $R_{ext} = A_0/A_1$ extrusion ratio, α die semi-angle
- The shear term grows with α , the friction term falls $\Rightarrow$ **optimum die angle** $\rightarrow$ optimisation problem

Didactic form — simplified (Avitzur-type upper bound for a conical die); see the full derivation in Hosford & Caddell and Avitzur before using the numbers.

Analytical methods Software

Solving analytical models: MATLAB · Python · GAMS

Problem type	Forming example	Tool
Closed form	Drawing force, Sachs stress	NumPy · MATLAB
Nonlinear equation	Neutral point in rolling	fzero · brentq
Differential equation	Slab-method forging, von Kármán	ode45 · solve_ivp
Symbolic computation	Deriving slab-method equations	SymPy · Symbolic TB
Optimisation <u>NLP</u>	Upper bound, optimum angle	fmincon · <u>GAMS/CONOPT</u>
<u>MINLP</u>	Pass schedule, number of passes	<u>GAMS/BARON</u> · Pyomo
Framework	Groove rolling	PyRoL (TU Freiberg)

```
# Upsetting force – slab method (Python)
import numpy as np
from scipy.integrate import quad
sf, mu = 250.0, 0.2 # MPa, -
R0, h0 = 10.0, 30.0 # mm
for h in np.linspace(30, 10, 5):
    R = R0*np.sqrt(h0/h) # volume constancy
    p = lambda r: sf*np.exp(2*mu*(R-r)/h)
    F = quad(lambda r: p(r)*2*np.pi*r, 0, R)[0]
    print(f'h={h:5.1f} mm F={F/1e3:7.1f} kN")
```

Why analytical methods are not enough

Aspect	Analytical methods	Numerical software (<u>FEM</u> / <u>FVM</u>)
Geometry	Simple (cylinder, strip, cone)	Arbitrary 3D from <u>CAD</u>
Material	Constant or mean $\underline{\sigma}_f$	$\underline{\sigma}_f(\varphi, \dot{\varphi}, T)$, anisotropy, damage
Output	Force, mean pressure	Full fields: strain, T, stress, defects, microstructure
Time to answer	Seconds	Minutes to days
Transparency	Every term is visible	"Black box" — must be checked

- Use analytical results for the **first estimate** and to **check FEM** results — leave the rest to the software

Part II — Forming Simulation Software

1 Ansys Forming · 2 Simufact Forming · 3 Abaqus · 4 Moldflow → 5 DEFORM (Part III)

Software Overview

Two groups of forming software

Dedicated process-oriented

- Templates: presses, tools, draw beads, passes
- For the **process engineer**, not the FEM specialist
- Sheet: **Ansys Forming**, AutoForm, PAM-STAMP
- Bulk: **Simufact**, **DEFORM**, QForm, FORGE
- Plastics: **Moldflow**, Moldex3D

General-purpose nonlinear FEM

- The user builds everything: parts, contacts, steps
- Maximum flexibility, user subroutines (UMAT)
- **Abaqus**, LS-DYNA, Ansys Mechanical, MSC Marc
- Research, teaching, unusual processes

Solver type	Principle	Where
Implicit (static)	Equilibrium in every increment, large steps	<u>DEFORM</u> , Abaqus/Standard, springback
Explicit (dynamic)	Small stable time steps, robust contact	<u>LS-DYNA</u> (Ansys Forming), Abaqus/Explicit
Finite volumes (Euler)	Fixed mesh, material flows through it	Simufact <u>EV</u> , Moldflow solvers

Every program follows the same workflow

1. **Geometry** — import from CAD (STEP, IGES, STL), symmetry



2. **Material** — flow curve, thermal data, damage



3. **Mesh** — element type, size, remeshing



4. **Boundary conditions** — friction, heat transfer, tool motion



5. **Solution** — solver, time steps, stopping criteria



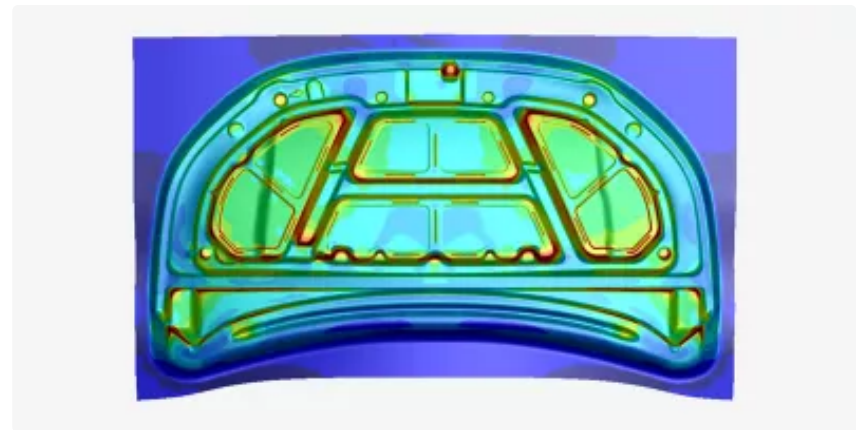
6. **Evaluation** — fields, curves, defects, report

- Steps 1–4 = **preprocessing** (lecture 4)
- Step 6 = **postprocessing** (lectures 5–6)
- Programs differ mainly in **how much of steps 1–4 their templates automate**
- "Garbage in, garbage out": the quality of the answer is set by the flow curve and the friction coefficient

1 · Ansys Forming Sheet metal stamping

Ansys Forming — stamping with the LS-DYNA solver

- Complete application for **sheet metal stamping**: pre-, solution and postprocessing in one environment
- Solver: LS-DYNA explicit (implicit for gravity and springback)
- Vendor: Ansys, part of **Synopsys** since 2025; first version 2022 R1
- Project types: **cold and hot stamping** (press hardening), hemming (beta), One Step (inverse method)
- For process engineers and tool makers — no need to be an FEM expert



Car hood inner panel in Ansys Forming (© Ansys / Synopsys)

1 · Ansys Forming Workflow · Results

Workflow and typical results

Process — CAD, tipping into the press, accuracy



Blank — blank outline, material, FLD, tailor-welded blanks



Operations — drawing, trimming, flanging, draw beads



Advance — springback compensation, trim line, robustness



Analysis — results, PPTX report

Result

Answers

FLD, formability index

Crack or wrinkle?

Thinning

Local thickness loss

Springback

Shape after unloading

Stoning, zebra

Visible surface defects

Draw-in, skid marks

Blank movement

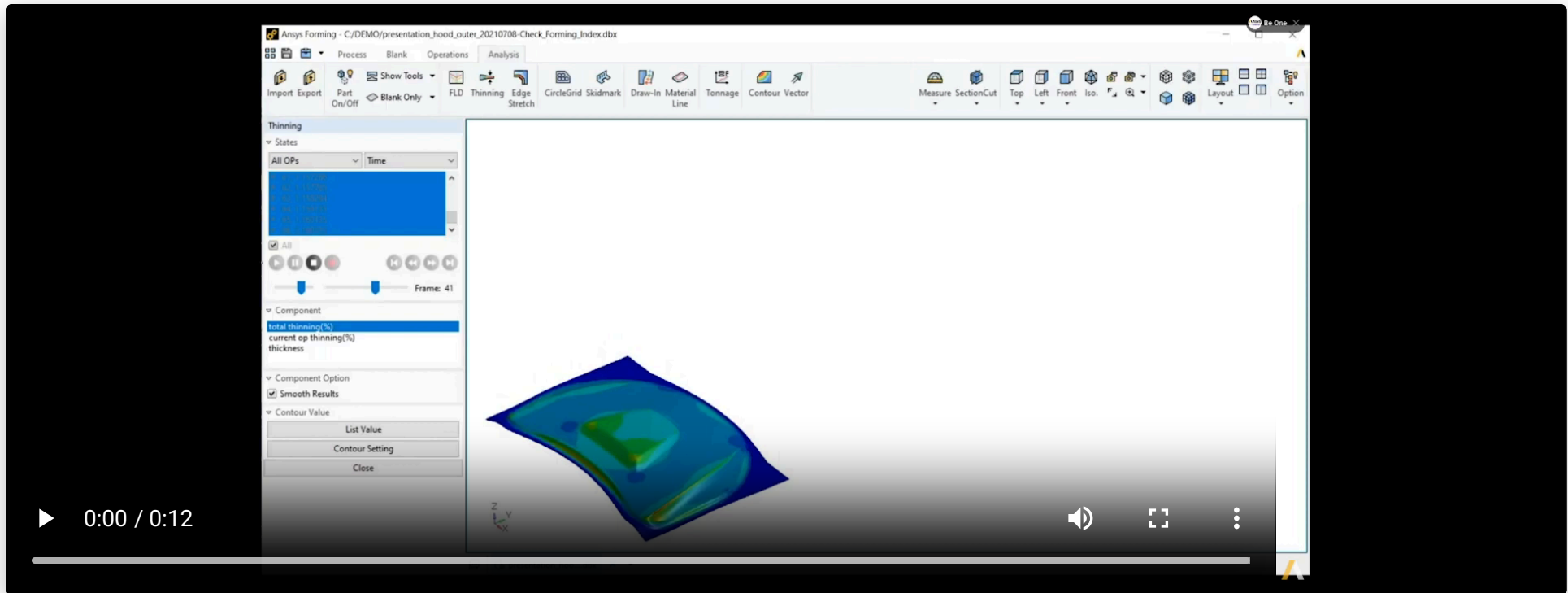
Press force

Press size

- Rule: **3 elements on a 90° radius**
- Presets: Fast · Accurate · Very Accurate

Source: Ansys Forming User's Guide 2026 R1

Video: Ansys Forming — stamping from start to finish



Ansys Forming — stamping simulation demo

2 · Simufact Forming Bulk · Sheet · Joining

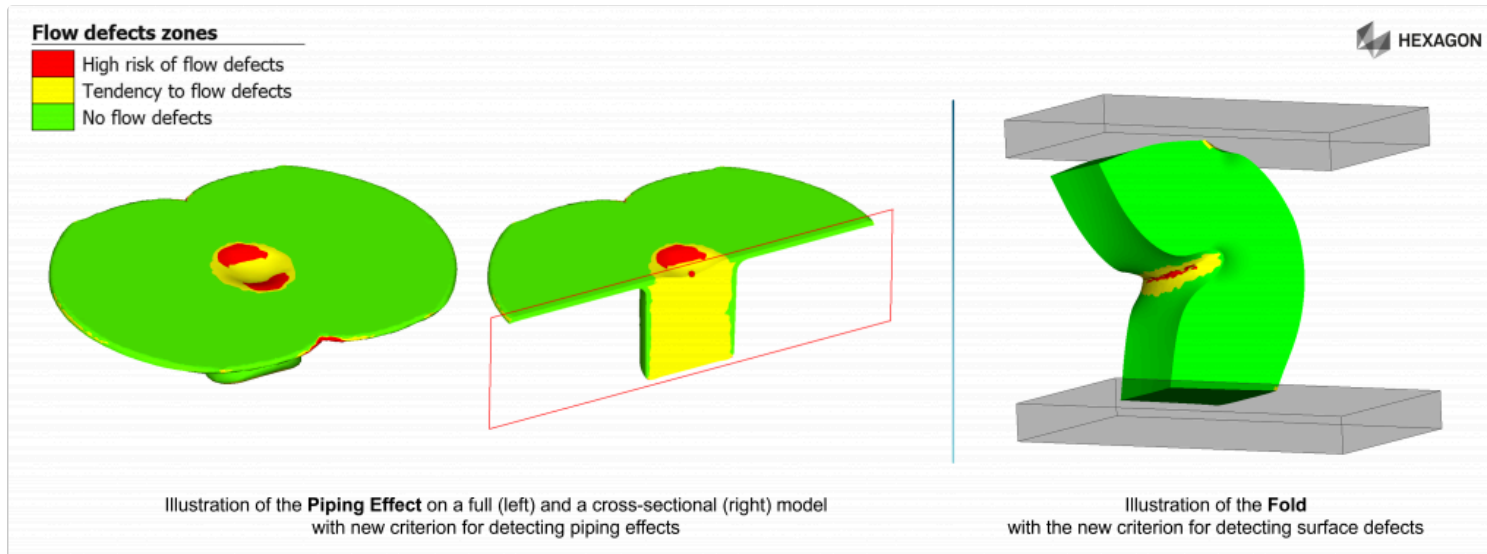
Simufact Forming — two solvers in one program

- Process-oriented software for **bulk and sheet metal forming**; Simufact (Hamburg) → MSC → Hexagon
- Forging, cold heading, extrusion, ring rolling, open-die forging, fine blanking, **self-pierce riveting**, **clinching**, heat treatment

	Finite elements (FE)	Finite volumes (FV)
Description	Lagrangian — the mesh moves with the material	Eulerian — fixed mesh, material flows through the cells
Mesh	Hexahedra, tetrahedra, remeshing	No remeshing needed
Suited for	Cold forming, sheet metal, tool stress, joining	3D hot closed-die forging with flash
Origin	MSC Marc solver	MSC SuperForge

- Switching FE ↔ FV between operations; FV parallelised in 2022 — up to **4× faster**

Simufact Forming — defect prediction



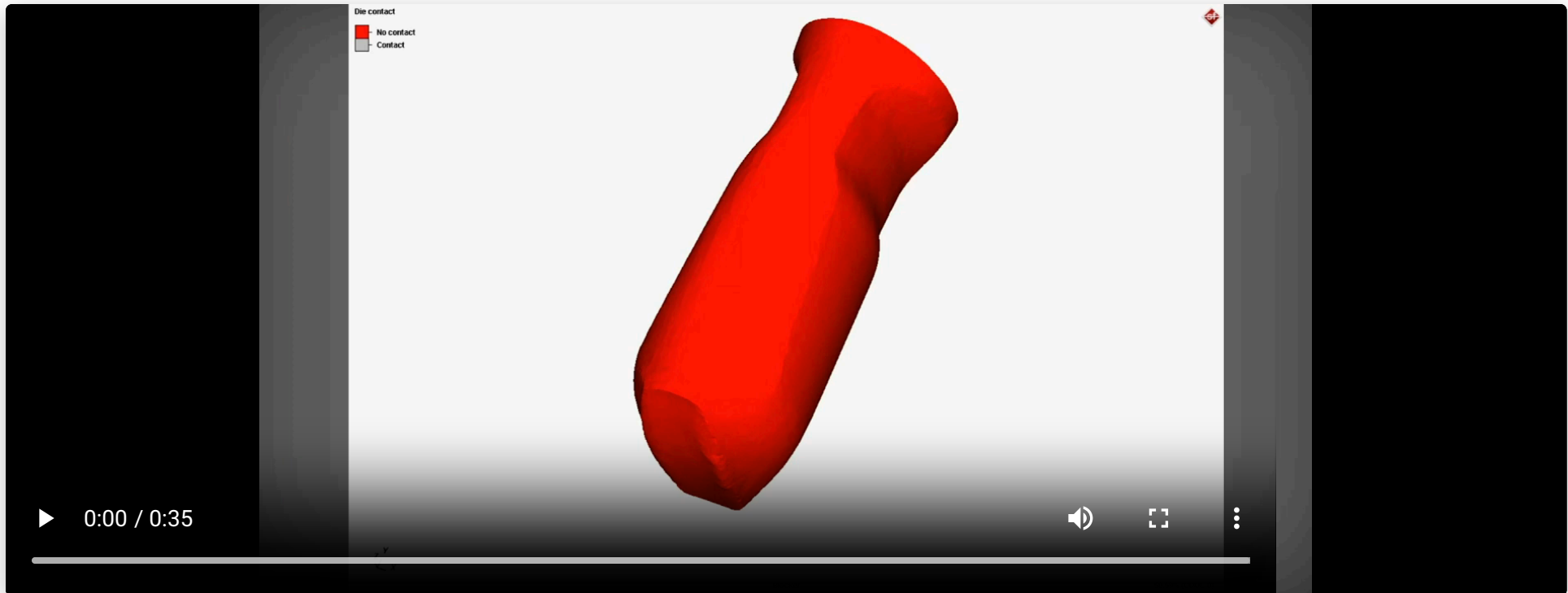
► Play animation in a new window

b) Damage-based crack initiation and propagation (© Hexagon)

a) "Flow defect zones": detection of piping and folds (© Hexagon)

- **Piping and folds:** typical in forging and extrusion — hard to predict without simulation
- Damage-based separation: 2D mesh splitting, 3D element deletion — blanking, fine blanking, riveting

Video: fold formation in forging a fork head



Simulation of folds: material flow during forging of a fork head (Simufact)

3 · Abaqus General-purpose nonlinear FEM

Abaqus — the general-purpose tool

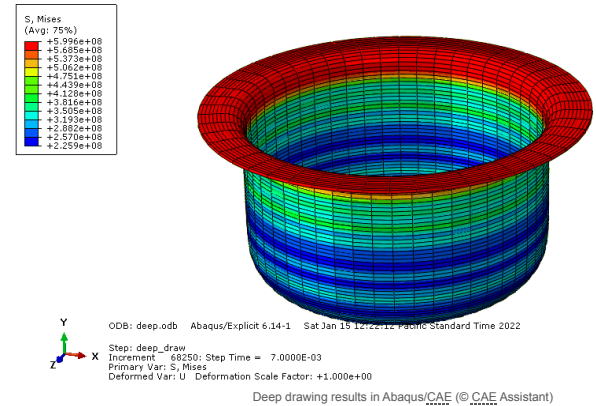
- Dassault Systèmes **SIMULIA**; Abaqus/CAE and 3DEXPERIENCE; free *Learning Edition* (≈ 1000 nodes)
- **No forming templates**: the user builds parts, contacts and steps — flexible, ideal for research (UMAT/VUMAT)

	Abaqus/Explicit	Abaqus/Standard
Use	Forming with complex contact, large deformations	Springback, smooth quasi-static loading, heat transfer
Strength	Robust contact, no convergence problems	True equilibrium, large increments
Watch out for	Small stable time step $\rightarrow$ mass scaling	Convergence with contact

Quasi-static check in Explicit: $\frac{E_{kin}}{E_{int}} = \frac{ALLKE}{ALLIE} \lesssim 5\text{--}10\%$

Abaqus example: axisymmetric deep drawing of a cup

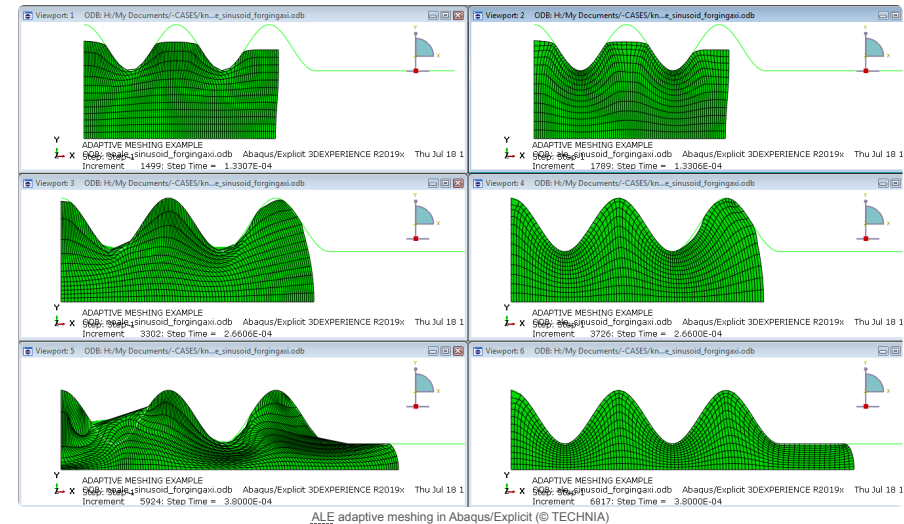
1. **Part** — blank: deformable axisymmetric shell; punch, die, blank holder: analytical rigid
2. **Property** — steel, $E = 200 \text{ GPa}$, $\nu = 0.3$, flow curve
3. **Step** — Dynamic, Explicit, 0.007 s
4. **Interaction** — penalty friction $\mu = 0.1$ (die, blank holder)
5. **Load** — punch 60 mm , **Smooth Step** amplitude
6. **Mesh** — blank only, refined at the radii
7. **Results** — PEEQ, thickness STH, punch force



Abaqus: large deformations and damage

Technique	Use
ALE adaptive meshing	Forging, rolling, extrusion (Explicit)
Adaptive remeshing	Error-driven (Standard)
Mesh-to-mesh mapping	Very large deformations, forging operations
CEL (Euler)	Extreme material flow

- Damage: ductile, **Johnson-Cook**, shear, **FLD**, **FLSD**, **MSFLD**, **M-K**
- Springback: $\geq 5-7$ integration points through the thickness, kinematic hardening



4 · Moldflow Plastic injection moulding

Autodesk Moldflow — forming of polymers

- Founded 1978 in Melbourne (Colin Austin); **Autodesk** since 2008; current version 2027
- **Adviser** (designers) · **Insight** (analysts) · Synergy interface · Communicator viewer
- Physics: **non-Newtonian viscous melt flow** + heat transfer — not solid plasticity
- Viscosity: Cross-WLF · PVT: Tait equation
- ≈ 13,900 thermoplastics in the database

Mould closing



Filling (injection) weld lines, air traps, short shot



Packing shrinkage, sink marks



Cooling largest part of the cycle



Ejection → **warpage**

Moldflow: mesh types and defects

Mesh	Principle	Use
Midplane 2.5D	Shells on the mid-surface; Hele-Shaw	Thin walls, fastest
Dual Domain 2.5D	Matched meshes on both surfaces	Thin simple parts
3D tetrahedra	Full 3D flow, no Hele-Shaw	Thick, massive parts
1D beam	Line elements	Runners and cooling channels

Defect	Remedy suggested by simulation
Short shot	Add/move gate, thicker wall, hotter melt
Weld lines	Move the gate to a non-critical area
Air traps	Venting, gate change
Sink marks	Core out thick sections, longer packing
Warping	Balance cooling, uniform wall

- Analysis sequence: **gate location** → **fill** → **pack** → **cool** → **warp** (+ fibre orientation → structural FEM)

Video: Autodesk Moldflow overview



Overview – Autodesk Moldflow (Autodesk)

Comparison of the five programs

Aspect	Ansys Forming	Simufact	Abaqus	Moldflow	DEFORM
Focus	Sheet metal stamping	Bulk, sheet, joining	General-purpose <u>FEM</u>	Plastic injection moulding	Bulk, <u>HT</u> , microstructure
Solver	<u>LS-DYNA</u> explicit	<u>FE</u> (Marc) + <u>FV</u>	Standard + Explicit	Flow (<u>2.5D</u> / 3D)	Implicit Lagrangian, <u>ALE</u>
Setup	Templates	Templates	Manual	Templates	Templates + open
Strength	Springback compensation	Fast <u>EV</u> hot forging	Flexibility, <u>UMAT</u>	Fill, pack, cool, warp	Remeshing, microstructure, wear
Vendor	Synopsys	Hexagon	Dassault Syst.	Autodesk	<u>SFTC</u> (USA)

- There is no "best" software — only software **suited to the process** and the question

Part III — DEFORM

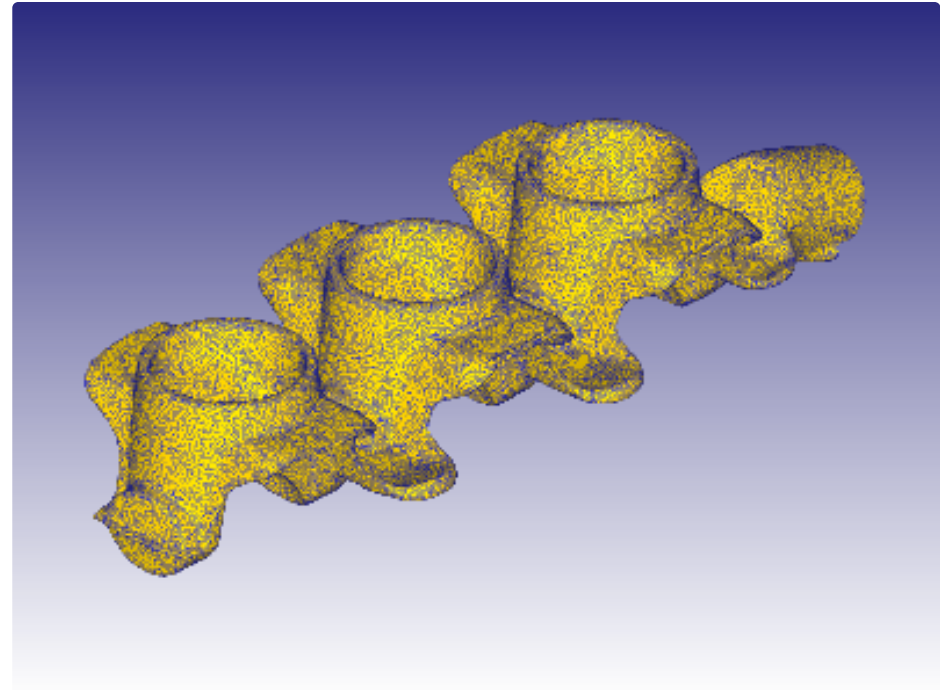
Design Environment for FORMing — the software of our lab sessions

5 · DEFORM Overview · History

DEFORM — from ALPID to a complete process simulator

- FEM system for **forming, heat treatment, machining and joining**
- Vendor: SFTC, Columbus, Ohio (≈ 30 people)
- Roots: ALPID at Battelle (U.S. Air Force) — the first practical FEM tool for production forming
- Theory: Kobayashi, Oh & Altan, *Metal Forming and the FEM* (1989)

Year	Milestone
1989	DEFORM-2D
1991	<u>SFTC</u> founded, takes over <u>DEFORM</u>
1993	DEFORM-3D
2024 · 2025	V14.0 · V14.1
2026	V14.1.1; V15.0 announced



Yoke forging — DEFORM-3D (© SFTC)

The DEFORM product family

Product	What it is
DEFORM-2D	Axisymmetric and plane problems — our first lab sessions
DEFORM-3D	Full 3D deformation, temperature and microstructure
DEFORM PREMIER	All systems and modules in one package
FORMING EXPRESS	Simplified system for forging, heading, extrusion, drawing
DEFORM-HT	Heat treatment: carburising, quenching, tempering
DEFORM-3D Machining	Turning, drilling, milling templates; tool wear (Usui)
Modules	Microstructure, DOE/optimisation, ring rolling, open-die forging, shape rolling, extrusion

- Processes: forging (hot/warm/cold), rolling, extrusion, drawing, heading, tube piercing, spinning, blanking, riveting, clinching, powder compaction, shot peening

The DEFORM solver: rigid-viscoplastic formulation

$$\pi = \int_V \bar{\sigma} \dot{\bar{\epsilon}} dV + \frac{K}{2} \int_V \dot{\bar{\epsilon}}^2 dV - \int_{S_F} F_i u_i dS \rightarrow \delta\pi = 0$$

- $\bar{\sigma}, \dot{\bar{\epsilon}}$ effective stress and strain rate · K penalty constant (incompressibility) · F_i surface tractions
- The unknowns are **velocities** → Newton-Raphson / direct iteration in every step (implicit, updated Lagrangian)
- Elastic strains are neglected — suited to **large plastic flow** (forging); elasto-plastic option for springback, residual stresses, tools
- ALE and steady-state solvers for ring rolling, shape rolling, extrusion, tube piercing

Source: Kobayashi, Oh & Altan (1989); DEFORM-3D brochure

5 · DEFORM Workflow

Working with DEFORM: Pre → Simulation → Post

DEFORM Preprocessor

objects, geometry, mesh, material, movement, inter-object conditions (friction, HTC), stopping, steps

↓ generate database (.KEY → .DB)

Simulation engine (FEM + automatic remeshing)

message and log files — watch convergence and remeshing



Postprocessor

state variables, force–stroke, point tracking, flow lines, damage

Object type	Meaning
Rigid	Non-deforming tools (optionally with heat transfer)
Plastic	Workpiece, rigid-viscoplastic
Elastic	Tool stress analysis
Elasto-plastic	Springback, residual stresses
Porous	Powder, sintered material

- Machine models: hydraulic, **mechanical**, **screw** press, **hammer** (energy)

Key inputs: flow stress, friction, heat transfer

$$\bar{\sigma} = f(\bar{\varepsilon}, \dot{\bar{\varepsilon}}, T)$$

- Tabulated data from the **material library** (steels, Al, Ti, superalloys, Cu ...) or from your own compression tests
- Neural-network prediction of flow stress from chemical composition (V13+)

$$\tau = m k = m \frac{\bar{\sigma}}{\sqrt{3}} \quad (\text{shear}) \quad \tau = \mu p \quad (\text{Coulomb})$$

Conditions	Friction factor m
Cold, good lubrication	0.08 – 0.12
Warm	≈ 0.2
Hot forging, lubricated	0.3
Hot forging, dry	0.7 – 1.0

- Die–workpiece heat transfer coefficient; the **Inverse HTC** module derives it from thermocouples
- **Most sensitive inputs:** flow stress and friction

m values: guide values for typical categories in [DEFORM](#)

Automatic remeshing and multiple operations

- The Lagrangian mesh deforms with the material → **automatic remeshing** when an element degenerates or penetrates the tool
- Tetrahedra (3D), quadrilaterals (2D), optionally hexahedra; **mesh windows** refine critical regions
- Each remeshing interpolates state variables → keep the number of remeshings reasonable (*volume loss*)

Multiple Operations (MO) — one chain, runs automatically

Furnace heating



Transfer (air cooling)



Forging blows 1...n



Flash trimming



Heat treatment → machining

5 · DEFORM Results

Results: from die filling to tool life

- Die filling, **laps and folds**, flow lines (point tracking, flow net)
- Effective strain, strain rate, temperature, triaxiality
- **Force–stroke** curve, energy → press selection
- Tool stress (elastic dies, prestressing) and **tool wear**
- ≈ two dozen **ductile damage** models, element deletion

Normalised Cockcroft–Latham criterion

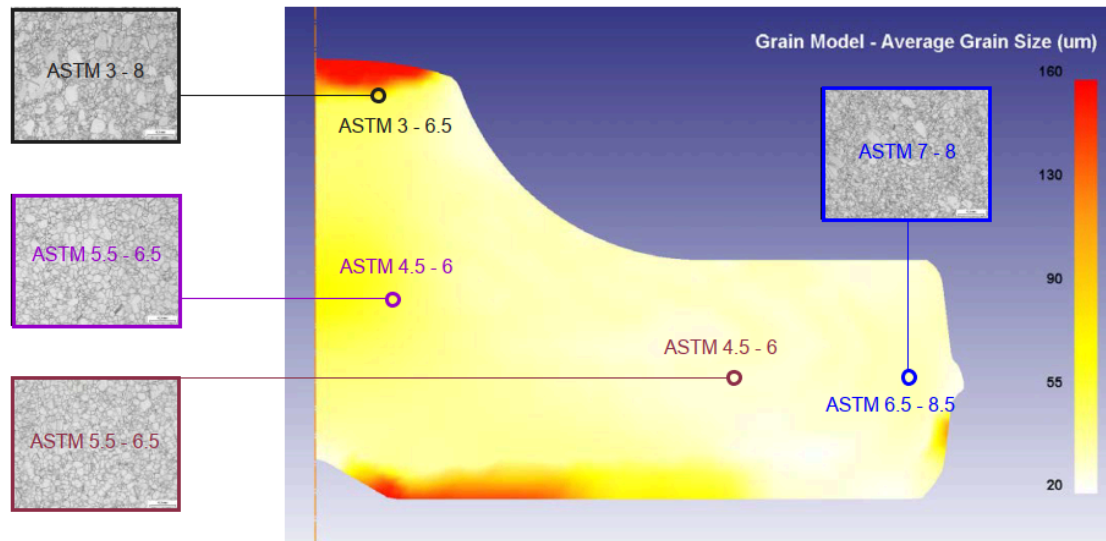
$$D = \int_0^{\bar{\epsilon}_f} \frac{\sigma_1}{\bar{\sigma}} d\bar{\epsilon} \geq C$$

Archard tool wear

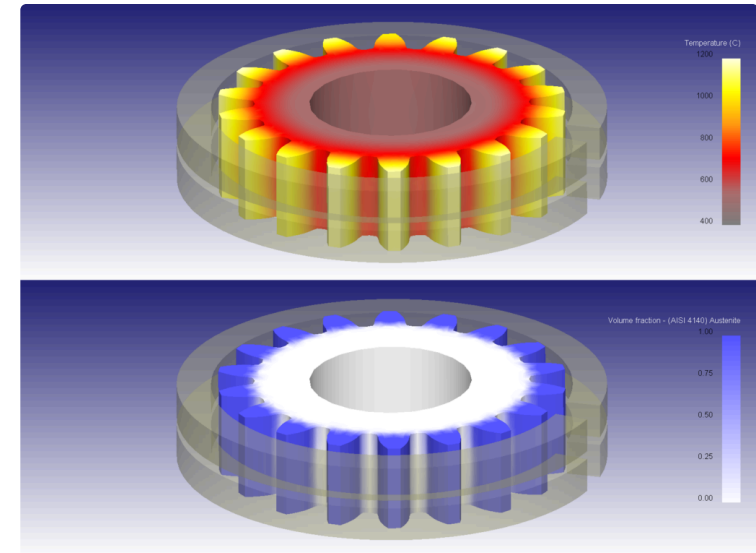
$$w = \int K \frac{p v}{H^a} dt$$

σ_1 maximum principal stress · p pressure · v sliding velocity · H tool hardness

Microstructure and heat treatment



a) Grain size in a forged Waspaloy turbine disc (© SFTC)



b) Induction-hardened spur gear (© SFTC)

$$\text{JMAK recrystallisation: } X = 1 - \exp \left[-\beta_d \left(\frac{\varepsilon - \varepsilon_c}{\varepsilon_{0.5}} \right)^{k_d} \right]$$

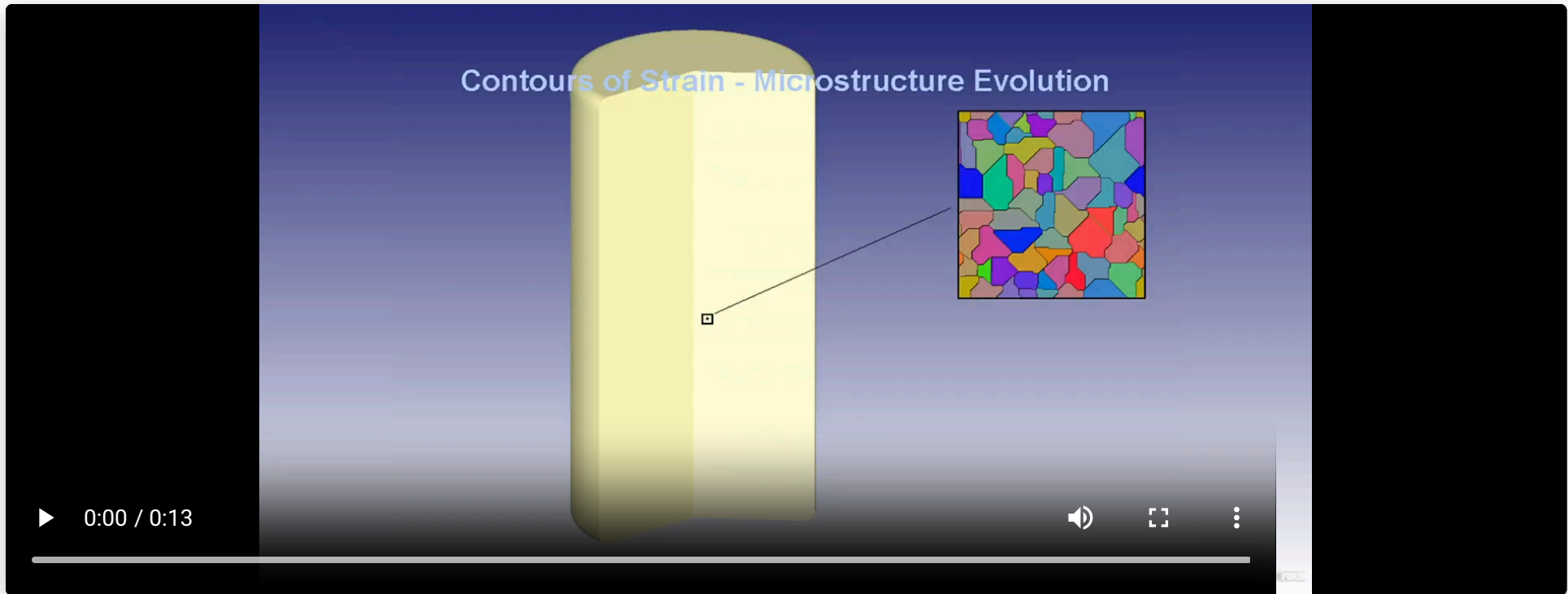
- Dynamic / static recrystallisation, grain growth, phase fractions, hardness, precipitates, texture (crystal plasticity)

Video: DEFORM — simulation of metal forming processes



DEFORM – The Premier Process Simulation Solution for Metal Forming (SFTC)

Video: forged turbine disc — microstructure simulation



Forged Turbine Disk Microstructure Simulation (SFTC)

DEFORM today: V14.1 → V15.0 (User Group Meeting, May 2026)

Area	New feature	Benefit
Solver	Stiffness reuse for elastic objects (<u>MUMPS</u> + <u>ESE</u>)	1.2 – 2.9× faster with elastic tools
Solver	Iterative elasto-plastic solver <u>EPCG</u> , <u>MPI</u>	> 50 % faster than <u>CG</u>
Spinning	<u>QEM</u> + partial domain	14 min instead of 17 – 35 h
AI	<u>U-Net</u> surrogate model for multi-pass shape rolling + <u>CMA-ES</u> groove optimisation	Design in seconds
Quenching	<u>CFD</u> boiling model (vapour film, nucleate boiling)	Realistic <u>HTC</u>
Blanking	3D mesh in the clearance zone and its remeshing	Fine mesh only in the clearance
Automation	Python DEFORM-API , <u>PPTX</u> report generation	Scripting, machine learning

Source: SFTC, DEFORM UGM Spring 2026 (12–13 May 2026)

DEFORM in our lab sessions

1. Upsetting a cylinder — **DEFORM-2D** axisymmetric



2. Compare force–stroke with the **slab method** (Part I)



3. Forward extrusion — die angle, friction, damage



4. Closed-die forging in **DEFORM-3D** — die filling, folds



5. Lab assignment (max. 20 points)

- Bring: your notes from Part I (the slab method formula)
- Check as you go: is the force realistic? was the volume preserved? how many remeshings were there?
- Assessment: assignment **20 pts** + written exam **80 pts**

Part IV — Special Simulation Software

Beyond Lagrangian FEM: inverse methods, finite volumes, particles, discrete elements, multiscale and data-driven tools

Special software Overview

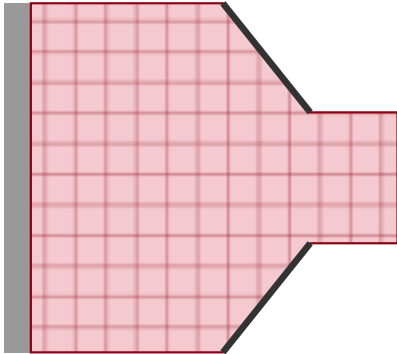
Methods and the software that implements them

Method	Principle	Software (examples)	Use in forming
Inverse one-step	Unfolding the finished part into a plane	AutoForm OneStep, Ansys Forming <u>One Step</u> , PAM-STAMP	Blank shape, quick feasibility
Finite volumes	Fixed Eulerian mesh	Simufact <u>EV</u> , QForm Extrusion*	Hot forging, AI extrusion
<u>SPH</u> (particles)	Meshless Lagrangian particles	DualSPPhysics, <u>LS-DYNA SPH</u>	Extreme deformations, <u>MIM</u> , <u>FSW</u>
<u>DEM</u>	Contact of discrete grains	EDEM, LIGGGHTS, Yade	Powder die filling
Multiscale	<u>CA</u> , crystal plasticity, phase field	DAMASK, <u>DEFORM</u> Microstructure	Recrystallisation, texture, earing
Data-driven	<u>ANN</u> / <u>ROM</u> trained on <u>FEM</u>	Python (PyTorch), DEFORM-API	Springback, rolling design
Analytical frameworks	Coupled analytical models	PyRoLL	Rolling pass design
Physical simulation	Reproducing ϕ , $\dot{\phi}$, T on a specimen	Gleeble	Flow curves, validation

* QForm uses FEM with coupled die deformation; listed as a dedicated extrusion tool

A simple example: one extrusion, four descriptions

Lagrangian FEM

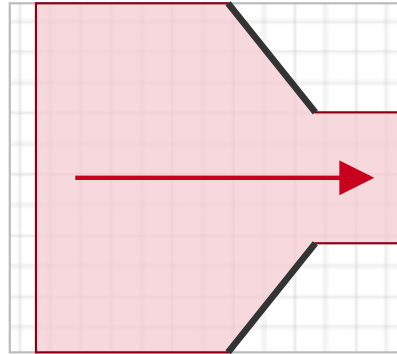


mesh moves with the material

→ remeshing needed

DEFORM, Abaqus, Simufact FE

Eulerian FVM

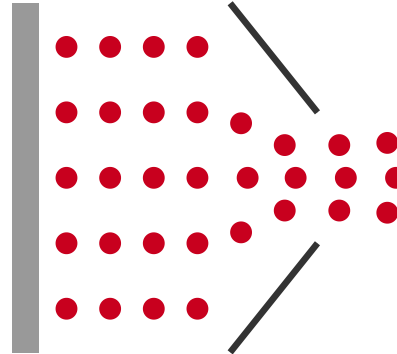


fixed mesh, material flows through

→ no remeshing

Simufact FV, Abaqus CEL

Particles SPH

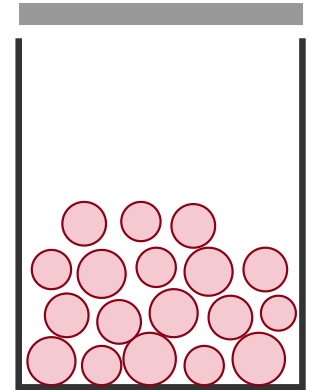


no mesh, particles carry history

→ extreme deformations

DualSPHysics, LS-DYNA SPH

Grains DEM



individual powder grains

→ filling, density

EDEM, LIGGGHTS, Yade

Schematic: how different methods discretise material flow through a die

Simple examples: inverse method and AI surrogate model

Inverse one-step (sheet metal)

Finished part geometry (CAD)

↓ unfold into a plane

Blank shape + strain estimate



Check in the FLD: risk of cracking / wrinkling

- Minutes instead of hours; ignores process history

Surrogate model (bulk forming)

Hundreds to thousands of FEM runs (DOE)

↓ training

Neural network (ANN, U-Net)



Instant prediction + optimiser

- Valid only within the trained design space

Special software Use cases

Use cases from research and industry

Problem	Approach	Result
Closed-die forging of a rib-web part	<u>UBET</u> + plasticine experiments	Force and flow agree with the physical model
Extrusion of Al profiles, ratio > 25	<u>FVM</u> on an Eulerian mesh (+ <u>ANN/GA</u>)	No remeshing; optimised die parameters
Metal powder injection moulding (17-4 PH)	<u>SPH</u> in DualSPHysics	Verified by real injections
Powder die filling (Distaloy AE)	<u>DEM</u> → density as input to compaction <u>FEM</u>	Filling shoe design before trials
Earing in deep drawing of <u>AA 5754</u>	Crystal plasticity <u>FEM</u>	Ears predicted from texture
Grain size in hot forging	Cellular automata coupled with DEFORM-2D	Recrystallised fraction and grain size
Springback in bending	<u>ANN</u> trained on <u>FEM</u>	Fast springback for new cases

Sources: literature review for this course (Kazan et al. 2009; Tikhovskiy et al. 2007; Fraunhofer IWM and others)

Video: dedicated extrusion simulation (QForm)

Simulation in QForm Extrusion Example 1

▶ 0:00 / 1:12



QForm Extrusion Software (QForm group) · more: [AutoForm channel](#) · [DualSPHysics](#) · [PyRoll](#)

Summary

- **Analytical methods** (slab method, upper bound, slip lines) give a quick force estimate and are the **first check** of every FEM result
- **Dedicated software** automates the process: Ansys Forming (stamping), Simufact (bulk + joining), Moldflow (polymers)
- **General-purpose FEM** (Abaqus) is flexible, but the user builds everything
- **DEFORM**: implicit rigid-viscoplastic FEM, automatic remeshing, damage, tool wear, microstructure — our tool for the lab sessions
- **Special software** goes beyond the limits of FEM: FVM, SPH, DEM, inverse and multiscale methods, AI surrogate models

NEXT TIME: MATERIAL MODELS — ELASTICITY, PLASTICITY, HARDENING, FLOW CURVES

Review questions

1. Why is the contact pressure in upsetting highest in the centre? What increases this peak? [answer](#)
2. Upper bound force: is it greater or smaller than the actual one — and why is that useful? [answer](#)
3. Explicit vs. implicit solver: which for stamping, which for springback? [answer](#)
4. What is the difference between the Lagrangian (EE) and the Eulerian (EV) description? [answer](#)
5. Why is the physics in Moldflow different from that in DEFORM? [answer](#)
6. Which two inputs in DEFORM affect the result most? What is the friction factor m ? [answer](#)
7. Name one damage model and one wear model in DEFORM and what they predict. [answer](#)
8. When would you use the inverse one-step method or an ANN surrogate instead of full FEM? [answer](#)

References and further reading

- Necpal, M.; Sobota, R.: *Modelovanie a simulácia procesov tvárnenia*. Spektrum STU, 2023 (in Slovak)
- Kobayashi, S.; Oh, S.I.; Altan, T.: *Metal Forming and the Finite Element Method*. Oxford UP, 1989
- Hosford, W.F.; Caddell, R.M.: *Metal Forming: Mechanics and Metallurgy*. Cambridge UP
- Dixit, P.M.; Dixit, U.S.: *Modeling of Metal Forming and Machining Processes*. Springer, 2010
- Bílik, J.; Kapustová, M.; Ridzoň, M.: *Teória tvárnenia*. AlumniPress, 2015 (in Slovak)
- SFTC: DEFORM brochures, DEFORM News 2022–2025, UGM Spring 2026 — deform.com
- Ansys Forming User's Guide 2026 R1 — ansys.synopsys.com
- Hexagon: Simufact Forming 2021.1 and 2022 — nexus.hexagon.com
- Abaqus Analysis User's Manual; TECHNIA and CAE Assistant tutorials
- Autodesk Moldflow help, What's New 2025–2027 — autodesk.com
- Weiner et al.: PyRoIL, JQSS 9(93), 2024

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